

Referee report on

“Trading social status for genetics in marriage markets: evidence from Great Britain and Norway”

In this paper, the authors study so-called social-genetic assortative mating (SGAM): the tendency that partners match on both socioeconomic status (SES) and genetics, causing the two to become correlated both within and across generations. This presents an alternative to “meritocratic social mobility” in explaining why inequality persists across generations. The authors set up a model which shows how genes-SES-gradients can appear because of SGAM even with no initial correlation. This also illustrates how the distribution of genetics is potentially endogenous to economic institutions or the environment more generally. The empirical part of the paper then tests the model. Specifically, it is shown that a more advantageous childhood environment due to lower birth order causes individuals to find partners with higher polygenic scores for educational attainment (PSEA). This effect appears to be mediated by the environment, in particular through education and, to some extent, income.

This is a very interesting paper, and quite different from other papers in the “genoeconomics” field that I am aware of. Maybe for that reason, I find it a bit difficult to judge how big of a contribution it makes. But I think this also comes down to the way the paper is written, in particular, the introduction could greatly benefit from spelling out the contribution (relative to existing literature) more clearly. A lot of emphasis is placed on the concept of SGAM. But it is already well-established that assortative mating is commonplace, both in relation to social outcomes like education as well as genetics. Thus, confirming the central argument – that both genes and SES are assets in marriage markets (and therefore become associated across generations) – does not seem that novel. Theoretically, is there anything in the model that is new compared to, e.g., Rustichini et al. (2023)?

Throughout the paper, SGAM is contrasted with meritocratic social mobility. I agree that the latter is probably the first channel that people have in mind when thinking of the persistence of social inequality. But I do not think anyone denies that assortative mating also plays a role. If this paper was showing that SGAM was really the *dominant* channel, I think that would have been a huge contribution. But although it is hinted at, it never actually attempts to test the two against each other.

The central contribution of this paper appears to be to show empirically that SES (birth order) has a causal effect on spouse genetics (and therefore on the genetics of the next generation). This also strikes me as a novel finding. Indeed, it is stated (at the very end of the introduction) that “*In particular, we are the first to show how SGAM interacts with institutional variables to affect economic inequality, mobility and associations between genes and SES, and the first to cleanly identify an environmental effect on spouse genetics*”. However, while the second part of the sentence is clear, I am a bit confused about the first contribution. But **unless I am missing something, the authors are not showing how SGAM interacts with institutional variables**. They are showing how it interacts with birth order and therefore also conceivably interacts with economic institutions (again, theoretically I am not sure how novel this is, but any empirical evidence probably would be). In any case, I suggest that the exact contribution is clarified and expanded on in the introduction.

In addition to more clearly spelling out the contribution, below are some other points that I think are important for the authors to think about:

- I generally found the model section quite difficult to read. I think it would benefit from some more meta-text (for example, the transition from the introduction is very abrupt). But I am also unsure whether it warrants more space. There are a lot of claims or propositions with little discussion of them, and many of them do not seem important for the rest of the paper, as the empirical part focuses on testing the initial assertion that attractiveness depends on both SES and genes ($0 < a < 1$). Again, I suggest that the central contribution from the theoretical part is clarified. Alternatively, moving the model (or parts of it) to the appendix might be preferable. Some smaller comments and questions regarding this part of the paper:
 - It is mentioned that adding a random component to attractiveness would reduce the effect of SGAM. Perhaps a more realistic extension would be to make the a -parameter individual-specific with a certain distribution, such that individuals are allowed to have different partner preferences. I imagine this would also retain the basic insights of the model while reducing the effects of SGAM, but this would be good to discuss briefly.
 - It is mentioned that individuals match if they are at the same quantile of attractiveness, but such quantiles do not seem to appear in the model. Are they meant to capture that the strength of SGAM can vary depending on the number of quantiles?
 - Are the A and C in the covariance matrix on page 9 related to the A and C in the ACE model? If so, please explain that in the paper.
 - Could you add some equations to make Claims 1-4 more transparent, or at least refer the reader to the proofs?
 - What is the meaning of the bars that appear over the variances in Proposition 3?
- The second model extension (letting genes influence SES) is obviously the much more realistic case, as parents affect their children's SES through investments in their skills etc., and this channel is influenced by genetics. But once you acknowledge this channel, it no longer makes sense to interpret θ narrowly as a policy variable ("economic institutions") either. SES is also transmitted across generations through parental investments and other social interactions. Thus, meritocratic mobility depends on both γ and θ . I would like to see some discussion of this, as I believe part of your claimed contribution rests on the narrow interpretation of this parameter.
- I am not very convinced about the mediation analysis. The decomposition rests on very strong independence assumptions which are unlikely to hold, even controlling for your observable characteristics. Educational attainment and income are likely influenced by a wide range of unobservable skills and preferences that also influence partner choice. Some of these

unobservables may initially be orthogonal to birth order, but controlling for educational attainment will then induce a correlation (collider bias) and change the birth order coefficient estimate. Similarly, educational attainment and income are almost certainly correlated with other mediators, which could also be other skills/preferences or other realized choices such as place of residence (the assumption that measured and unmeasured mediators are uncorrelated is implausible). A plausibly causal mediation analysis is difficult without either an instrument or some very persuasive robustness checks. I believe **you need to be a lot more up front in acknowledging this limitation** and interpret these results as more suggestive unless you can improve their credibility.

- It is somewhat concerning to me that controlling for parental age affects the birth order estimates so much. My experience is that this usually matters quite little, and it makes me wonder whether this control induces some kind of selection. I am not sure exactly what it would be, but one could imagine that holding parental age at birth fixed while varying birth order would imply some other change in the type of families being compared (e.g., it would probably change the year of birth, although you control for this). Could you do anything to investigate this further?
- Relatedly, one potential issue is that your birth order estimates rely on between-family variation rather than comparing siblings within the same family. I suggest two things to check the robustness of these results: (1) You could estimate the family-type fixed-effects specification from Black et al. (2018). (2) You could also estimate your main specification using the smaller sibling sample that you use to test that $\alpha > 0$ (Table 9), adding family fixed effects – then, both birth order and the PSEA would be clearly exogenous.
- Your sample size often varies a lot across columns within the same tables. Thus, many of the comparisons you are making are between very different samples. I would like to see these being done holding the sample fixed (e.g., the main calculation regarding how much of the birth order effect that is explained by education, but also other comparisons).
- What happens if you only control for education/income without the other mediators (height etc.)? I wonder if they make a difference. I also think the comparison between the two countries would be cleaner in this case given that the set of controls is different.
- While the apparent difference in the importance of income in the two countries is interesting, I worry that a lot of it comes down to the measures being very different. In the UK data, you are measuring expected income based on occupation. You mention that the household income measure is inappropriate partly because it is measured after marriage. But this could also be the case in the Norwegian data where it is measured at age 30. Thus, the apparent lack of importance of income in Norway could reflect some kind of reverse causality. Could

you homogenize the measures, for example by basing the Norwegian measure also on occupation? If not, I think you need to be clearer about this limitation.

Below are some additional minor points:

- The wording in the introduction that genetic inheritance guarantee that birth order is independent of genetic endowments comes off as too strong given the potential for selective fertility (which you do discuss, but only later).
- Relatedly, are the birth order estimates sensitive to not controlling for the PSEA?
- One of the citations on page 21 is broken.
- Maybe it is just me, but I find it confusing that you apply the Bonferroni correction when doing the balancing/placebo tests. Why not just report how many of the coefficients are significant at conventional levels and see if it is close to what would be expected by chance?
- Why are almost all the placebo tests done using the UK data? MoBa is only mentioned at the end of footnote 23. Could you expand a bit on these tests?
- Why don't you estimate the results by separate family size (Figure 6) using MoBa which is much more precise?
- When discussing the results in Table 8, could you help the reader by clarifying which estimates refer to which direction (effect of male birth order on female PSEA or vice versa)?

References:

Black, Sandra E., Erik Grönqvist, and Björn Öckert. "Born to lead? The effect of birth order on non-cognitive abilities." *Review of Economics and Statistics* 100.2 (2018): 274-286.

Rustichini, Aldo, et al. "Educational attainment and intergenerational mobility: A polygenic score analysis." *Journal of Political Economy* 131.10 (2023): 2724-2779.